

Practical 8

Internet of Things (IoT)

Learning Objectives

Upon completion of this practical, you will be able to:

- Prototype basic connected IoT devices.

Scenario

- a) **Lab Activity 1:** Understand how IOT plays a part in the world.
- b) **Lab Activity 2:** Explore and modify a smart home.
- c) **Lab Activity 3:** Program a simple IOT device.

Lab Activity 1

In this activity you will work to understand the extent that IOT devices are currently embedded into your home, and university campus.

Activity Objectives:

- Obtain a basic understanding of what the IOT is.
- List current smart devices and things in your home, office and campus.

Task 1: Obtain a basic understanding of what IOT is

There are numerous definitions for what should and should not be included as part of the internet of things, but in a general sense it refers to the billions of connected objects and devices which make up our world. According to Cisco, over 50 billion devices are expected to be connected over the internet by 2020, and over 500 billion things are expected to be connected to the Internet by 2030.

A thing in the Internet of things is anything that is accessible over a network. And can include:

- Simple objects, such as smart lightbulbs.
- Sensors, such as temperature and motion sensors.
- Living organisms, such as animals tagged with biochip transponders.
- Complex devices, such as smart fridges or smart phones.

As sensors and embedded systems continue to grow cheaper more things are becoming internet enabled. As more objects get connected, this fundamentally changes the interactions between the physical and digital world, bringing new business opportunities, new security threats, new automation opportunities, new challenges and new data.

Task 1: List current smart devices and things in a home, office and campus

a) List some connected things you may find in a smart home.

b) _____
List some connected things you may find in a smart office?

c) _____
List some connected things you may find in a smart university campus.

d) _____
List some connected things you may find in a smart city?

Lab Activity 2

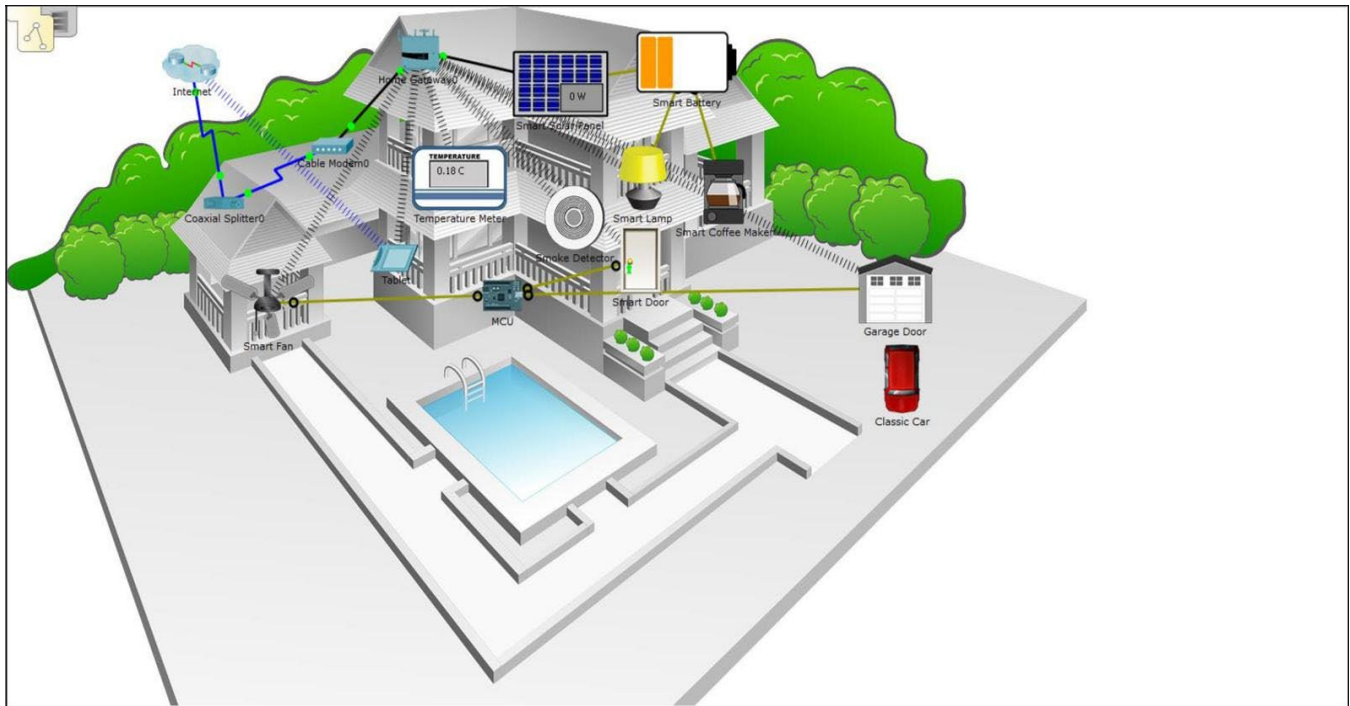
In this activity you will open a Packet Tracer file with an existing home network, explore the devices on the network and then add additional wired and wireless IoT devices.

Activity Objectives:

- Explore the Existing Smart Home Network
- Add Wired IoT Devices to the Smart Home Network
- Add Wireless IoT Devices to the Smart Home Network

Activity 2 – Adding IoT Devices to a Smart Home

The Smart Home Network



Part 1: Explore the Existing Smart Home Network

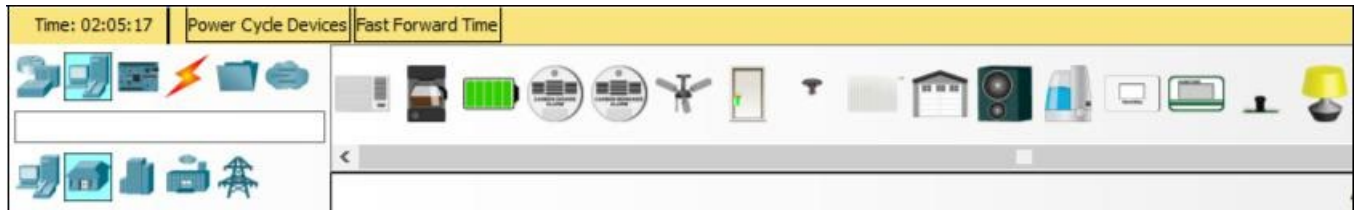
Step 1: Open the Smart_Home_Network.pkt file

- Download and open the **Smart_Home_Network.pkt** file.

Step 2: Explore the Smart Home Network

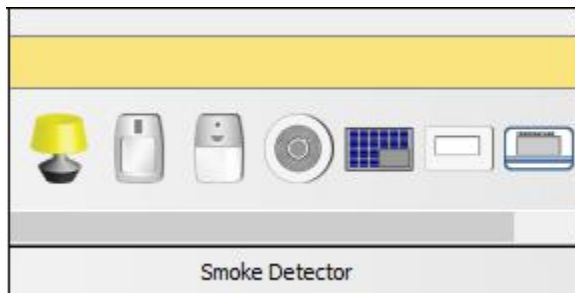
a) Explore IoT end devices.

At the bottom left corner of the Packet Tracer window, locate and click the **[End Devices]** icon in the top row, and the **[Home]** icon in the bottom row of the **Device-Type Selection** box.

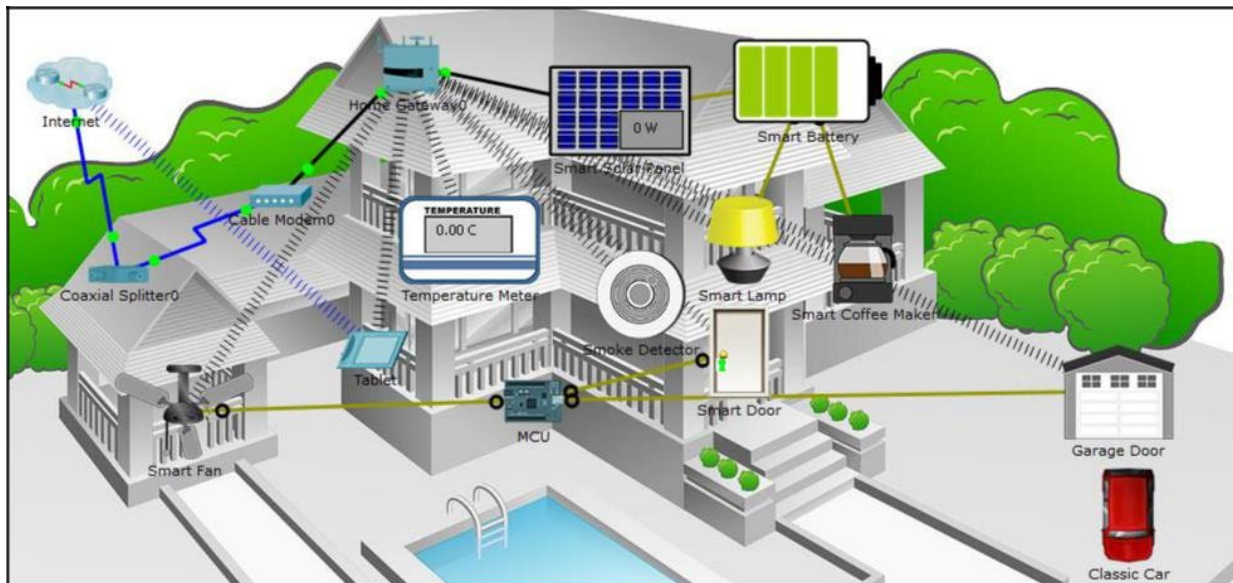


Across the bottom of the Packet Tracer window, the **Device-Specific Selection** box displays the many different Smart Home IoT devices available.

Move the mouse pointer over each device and notice that the descriptive name of the device is displayed at the bottom of the **Device-Specific Selection** box. Take a moment to look at each device type.

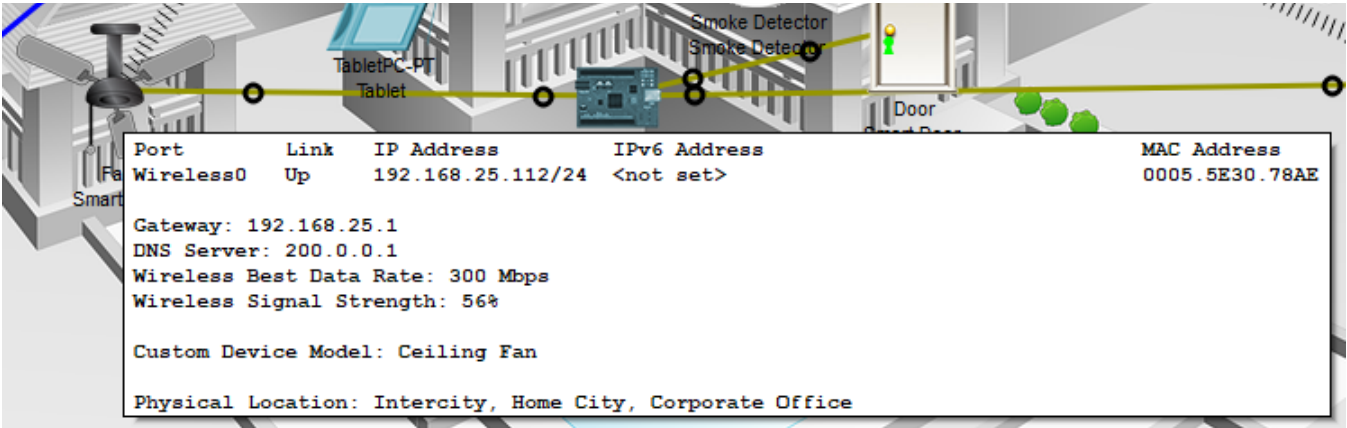


b. Explore the Smart Home network.



In the Logical workspace is a pre-built smart home network that consists of many wired and wireless IoT devices, and network infrastructure devices.

When you place your curser over a device, such as the Smart Fan, an informational window opens containing basic network information about that device.



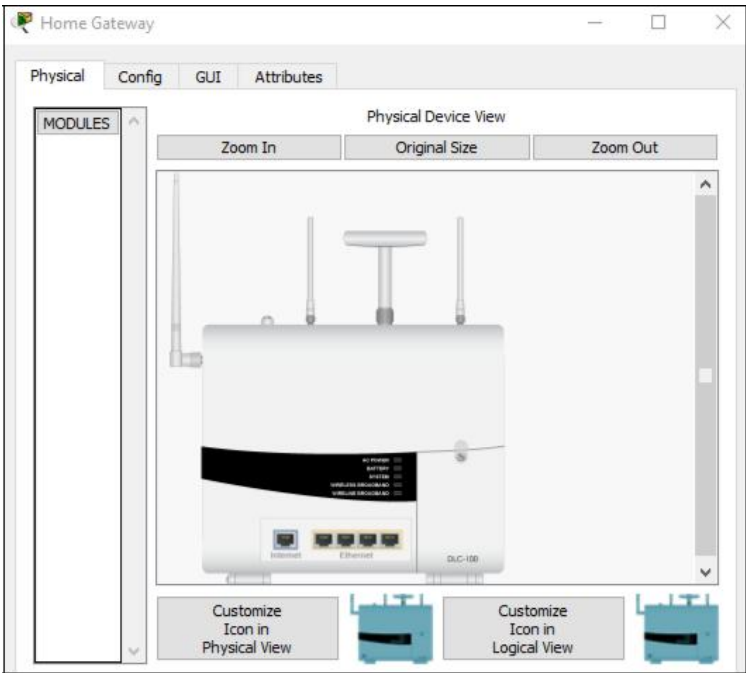
To turn on or activate a device, simply hold down the **Alt** key on the keyboard and then **left-click** the device. Try this on each of the smart devices to observe what they do.

The smart home network also consists of infrastructure devices such as a home gateway.

Click the **Home Gateway** icon to open the Home Gateway window.

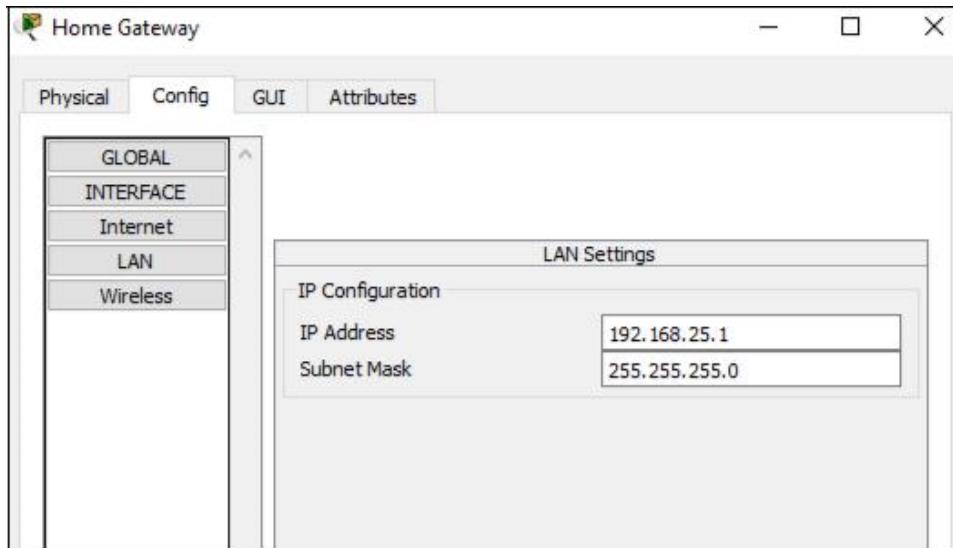


The Physical tab is selected by default and shows a picture of the Home Gateway.



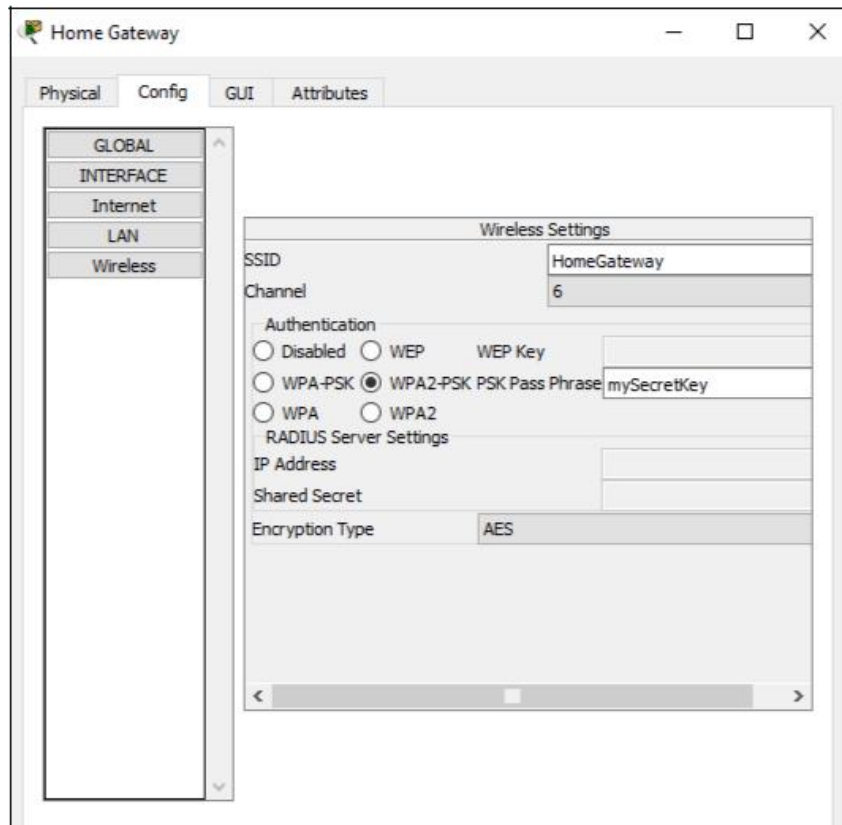
Next, click the **Config** tab and then in the left pane click **LAN** to view the LAN Settings of the Home Gateway.

Write down the IP Address of the home network for future reference. _____



Click **Wireless** in the left pane to view the wireless settings of the Home Gateway.

Write down the SSID of the home network _____ and the WPA2-PSK Pass Phrase _____ for future reference.

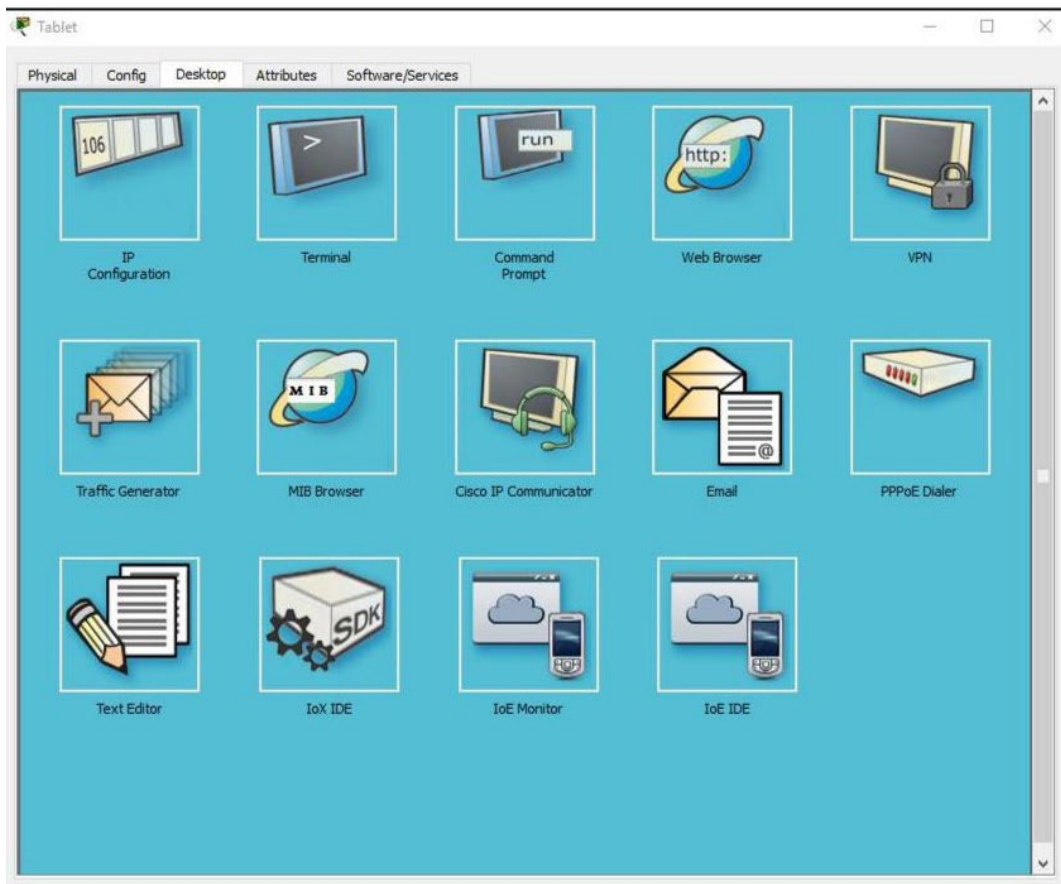


Close the Home Gateway window.

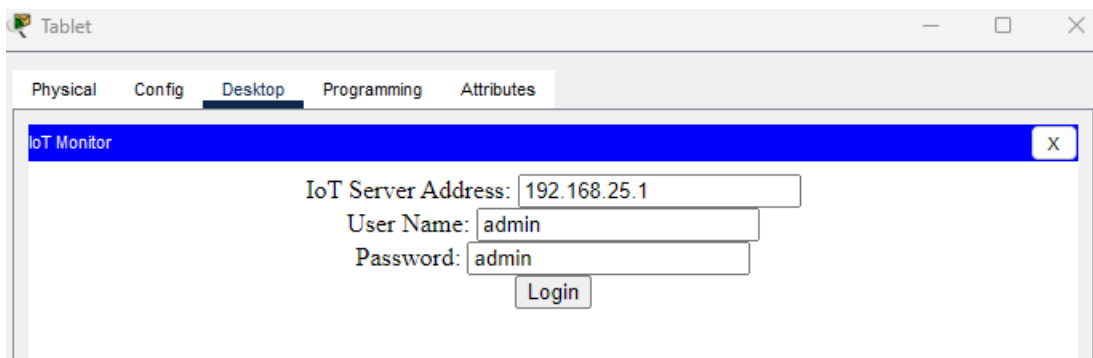
Next, click the **Tablet** device icon to open the Tablet window.



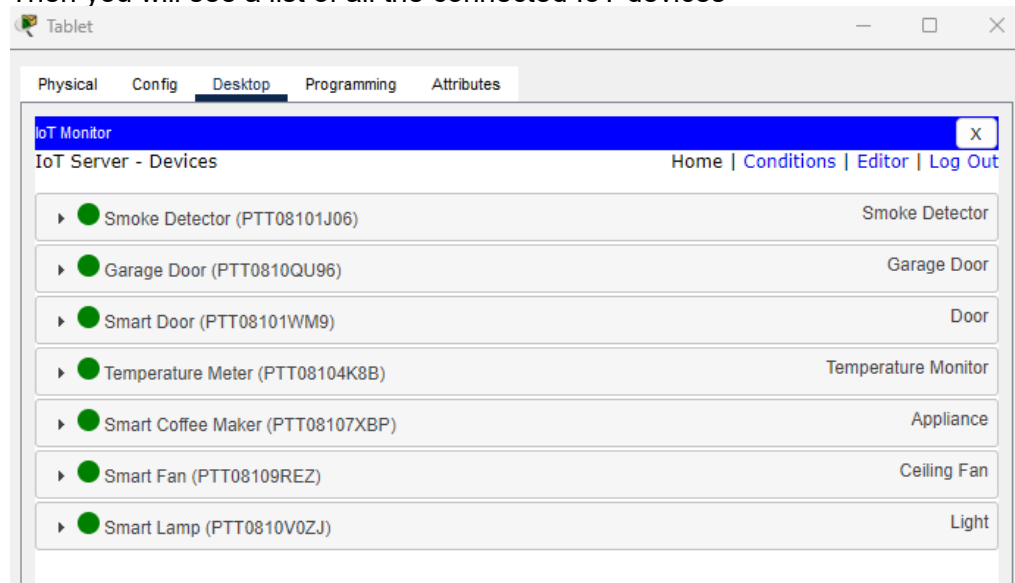
In the Tablet window, select the **Desktop** tab and then click the **IoT Monitor** icon.



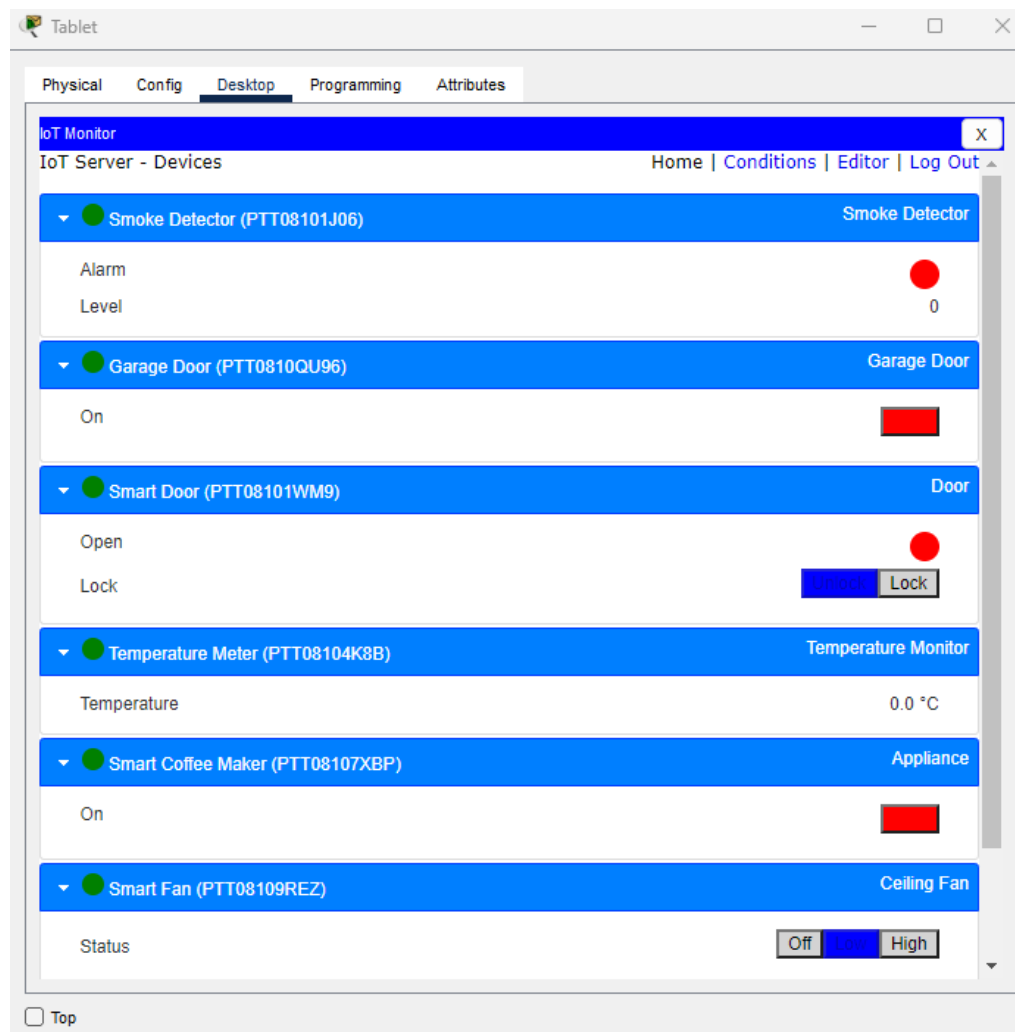
In the **IoT Monitor**, type **192.168.25.1** into the IoT Server Address box, type **admin** for both the username and the password and click **Login**.



Then you will see a list of all the connected IoT devices



Click a device in the list, the status and settings of that device is displayed.

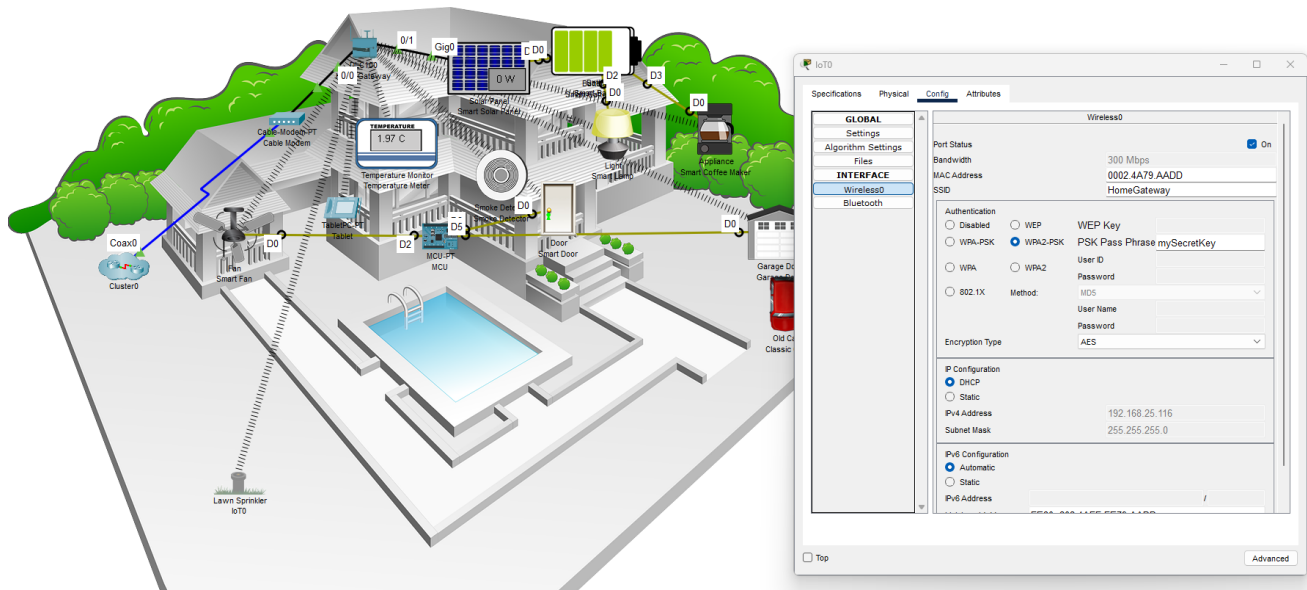


Close the Tablet window.

Part 2: Add Wired IoT Devices to the Smart Home Network

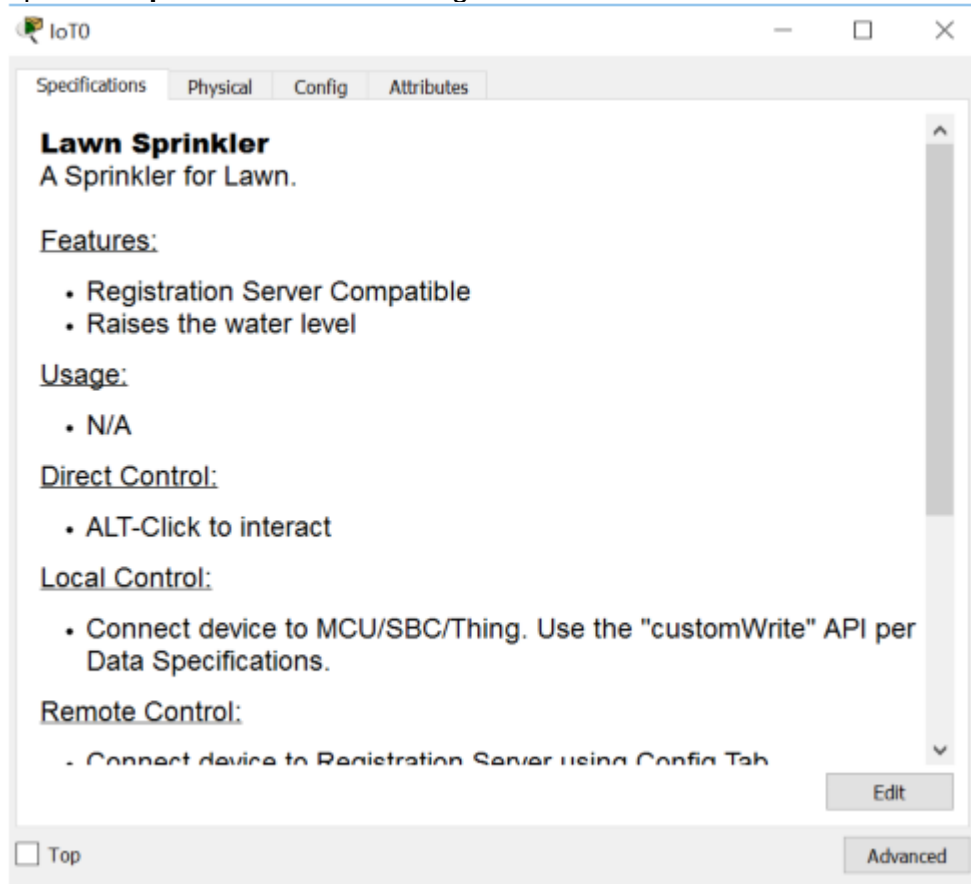
Step 1: Set up a device to the wireless network

- In the Device-Specific Selection box, click the Lawn Sprinkler icon and then click in the workspace where you would like to locate the Lawn Sprinkler (IoT)
- Set up the wireless connection the Lawan Sprinkler to the Home Gateway. Click the **Lawn Sprinkler** device icon in the workspace and enter PSK Pass phrase **mySecretKey**



Step 2: Configure the sprinkler for network connectivity

Open the **Specification** tab which gives information about the device which can be edited.



- a) Click the **Config** tab to edit the device configuration settings. In the Config tab, make the following changes to **Settings**:
- Set the **Display Name** to Sprinkler1 (notice the window name changes)
 - Set the IoT Server to **Home Gateway**

The screenshot shows the 'Sprinkler1' configuration window with the 'Config' tab selected. The left sidebar has a tree view with 'GLOBAL' expanded, showing 'Settings', 'Algorithm Settings', and 'Files'. Under 'INTERFACE', 'Wireless0' is selected. The main area displays 'Global Settings' with the following fields:

- Display Name: Sprinkler1
- Serial Number: PTT08108JUX-
- Interfaces: Wireless0 (dropdown)
- Gateway/DNS IPv4:
 - ☒ DHCP
 - ☐ Static
 - Default Gateway: 192.168.25.1
 - DNS Server: 200.0.0.1
- Gateway/DNS IPv6:
 - ☒ Automatic
 - ☐ Static
 - Default Gateway: (empty)
 - DNS Server: (empty)
- IoT Server:
 - ☐ None
 - ☒ Home Gateway
 - ☐ Remote Server
 - Server Address: (empty)

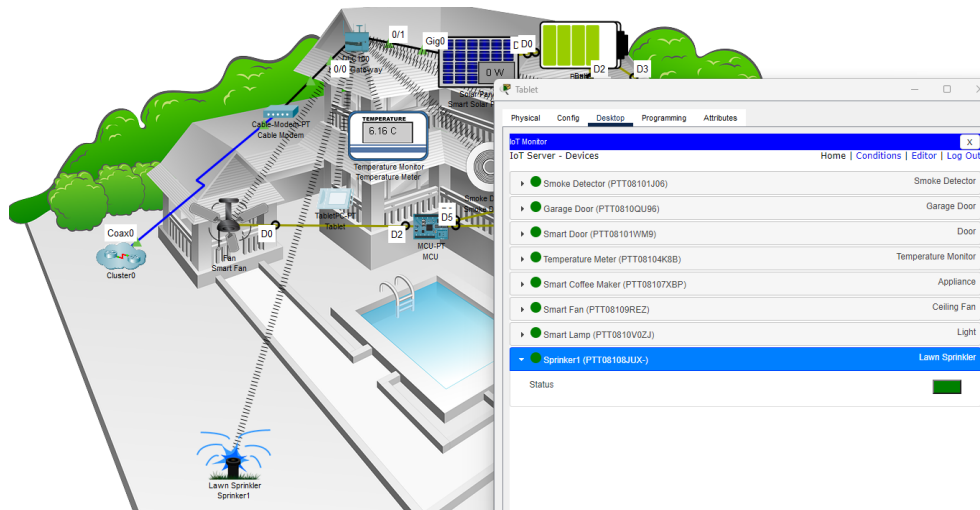
At the bottom left is a 'Top' button and at the bottom right is an 'Advanced' button.

Close the **Sprinkler1** window.

- b) Verify that the sprinkler is on the network.
Go to the IoT Monitor from the **Tablet**.

The screenshot shows the 'Tablet' application window with the 'Desktop' tab selected. A blue bar at the top is labeled 'IoT Monitor'. Below it, the login form contains the following fields and buttons:

- IoT Server Address: 192.168.25.1
- User Name: admin
- Password: admin
- Login button



The device Sprinkler 1 should now appear in the IoT Server – Devices list. You can switch on the Sprinkler1

Close the Tablet window.

Lab Activity 3:

Blockly is a visual programming language that lets users create programs by connecting blocks, that represent different logic language structures, rather than by writing the actual code. Blockly runs within a web browser and can translate the visually created program as JavaScript, PHP, or Python. In this lab, you will use Blockly to examine Blockly programming and to control an LED.

Activity Objectives:

- Learn how to program MCUs using blocky
- Control a RGB LED in packet tracer using blocky
- Program an MCU using JavaScript

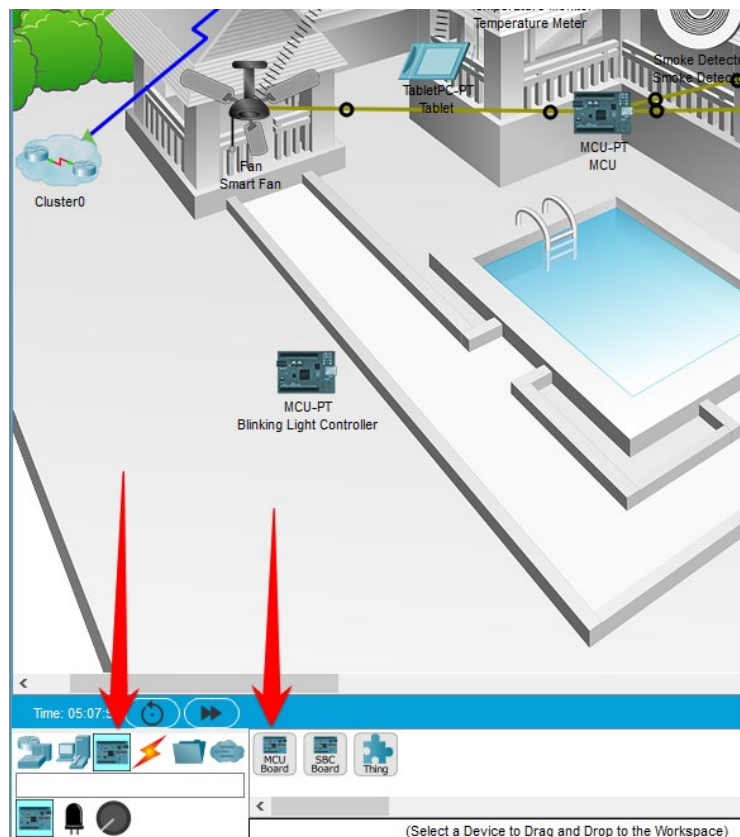
Required Resources:

Cisco Packet Tracer 7.1.1 and above is installed and available.

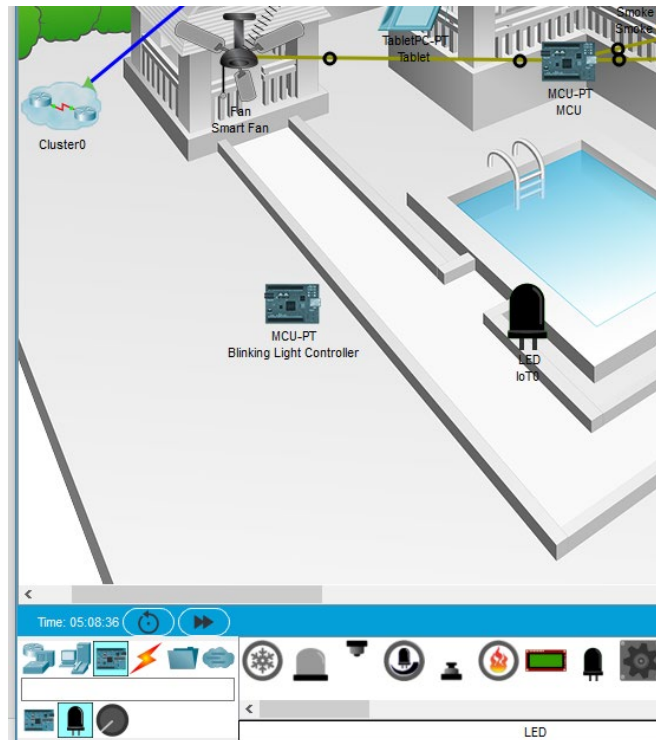
Task 1: Learn how to program MCUs using blocky

In Part 1, you will access the Cisco Packet Tracer program and examine LED control using Blockly programming.

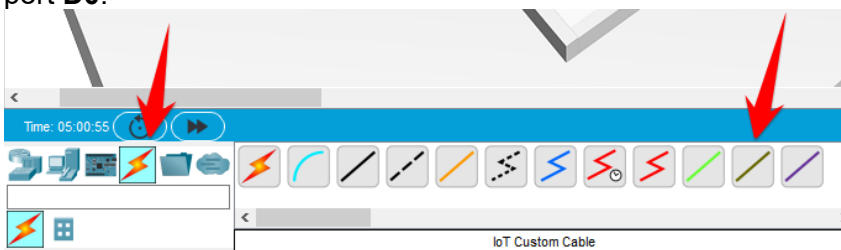
- a) Click on the **Components** category, then click **MCU Board** and drag it to the workspace.



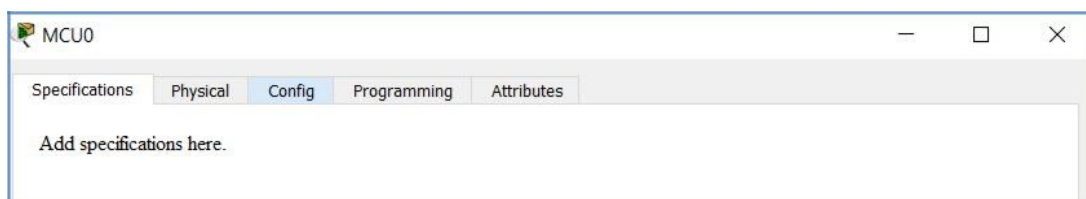
b) Click the subcategory **Actuators**, select **LED** and drag it to the workspace.



c) Click the **Connections** category, select **IoT Custom Cable** to link MCU at port **D1** and LED at port **D0**.



c) Double click the **MCU**. Its configuration window displays.

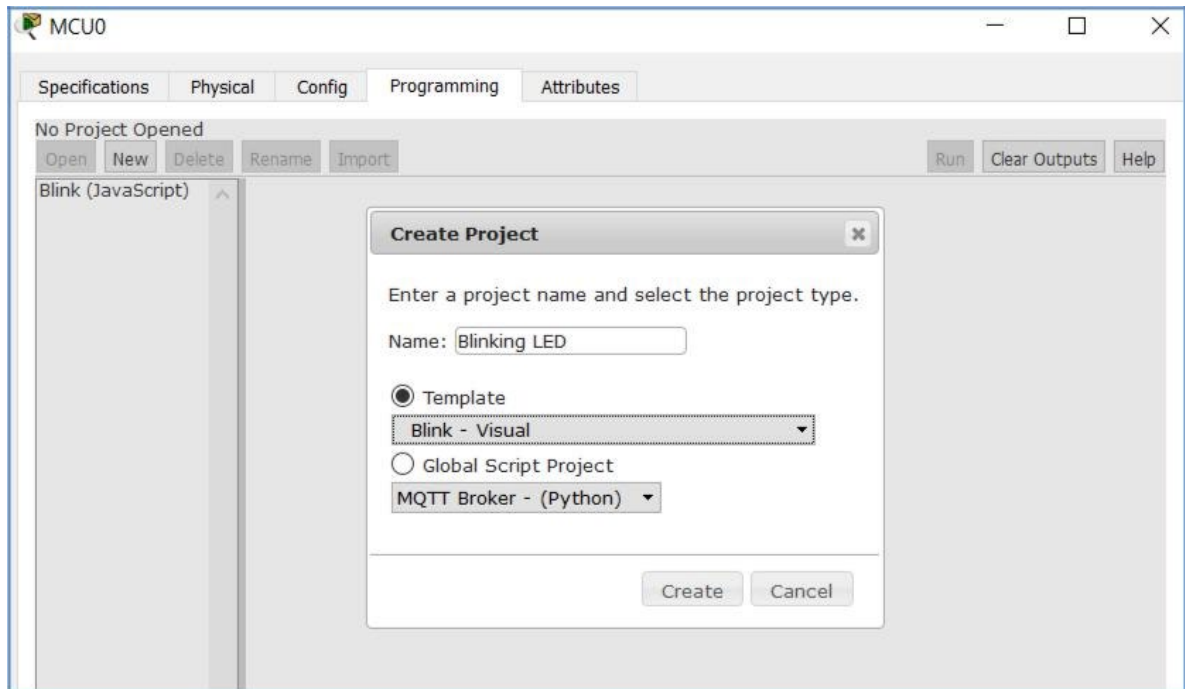


d) Click the **Programming** tab. (If you do not see the Programming tab, click the **Advanced** button at the lower right corner.)

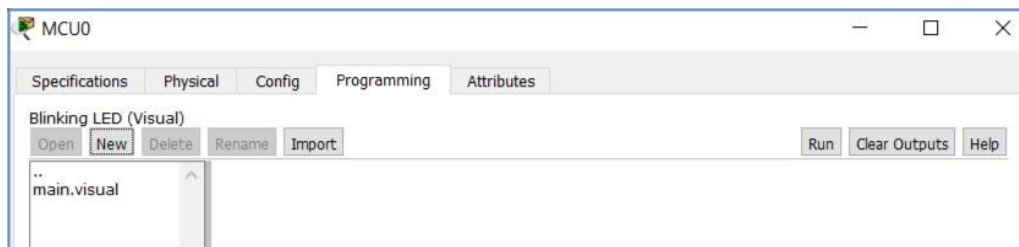


Step 2: Examine a prebuilt Blockly program

- a) Under the note **No Project Opened**, click **New**. In the **Create Project** window, enter **Blinking LED** as the project name. In the **Template** dropdown menu, select **Blink – Visual**.



- b) Click **Create**.

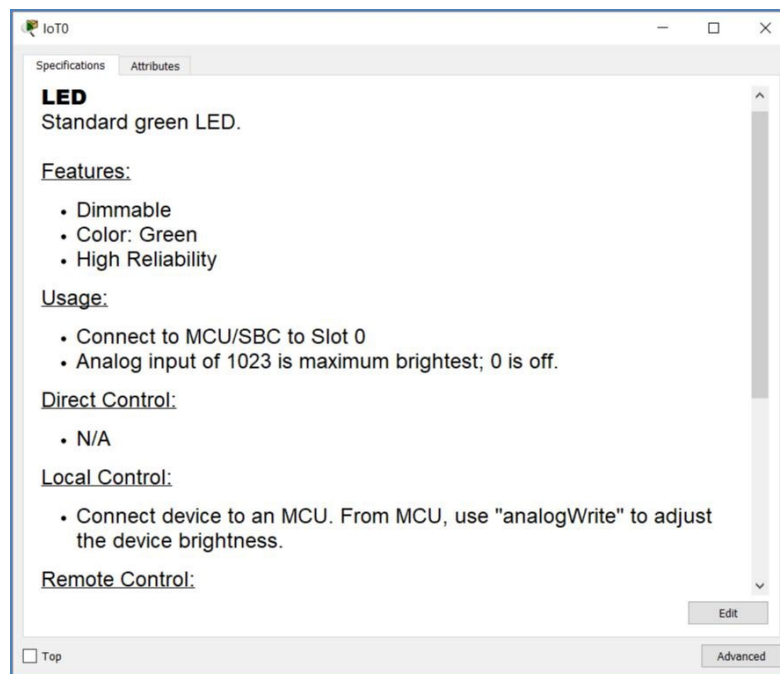


- c) Double-click **main.visual**. The prebuilt Blockly program displays.

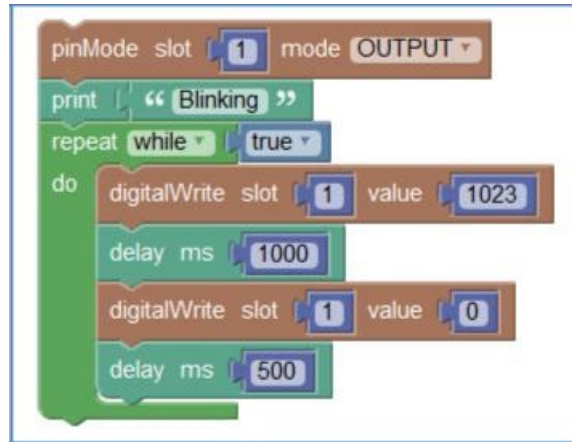


d) Click **Run**. Does the LED blink?

e) Click on the LED and study its specification.



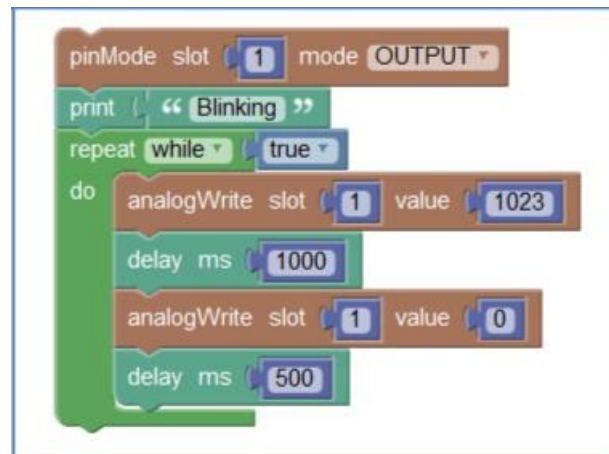
f) Click **Stop**, and change the **Value** field of the first **digitalWrite** block to 1023 (the Analog input of the LED for maximum brightness listed under Usage). Change the **Slot** field of both digital write blocks to what is listed under the Usage section of the LED specifications. In the above example, this value is 0.



g) Click **Run**, does the LED blink?

Why was it not blinking before? What about setting the value 500 or 800?

h) Looking at the LED specifications again, it indicates that we can use “analogWrite” to adjust the device brightness. Expand **Pin Access** group and use the **analogWrite** block to replace **digitalWrite** block.



i) Now change the values of the first and second **analogWrite** blocks and observe the different LED brightness levels.

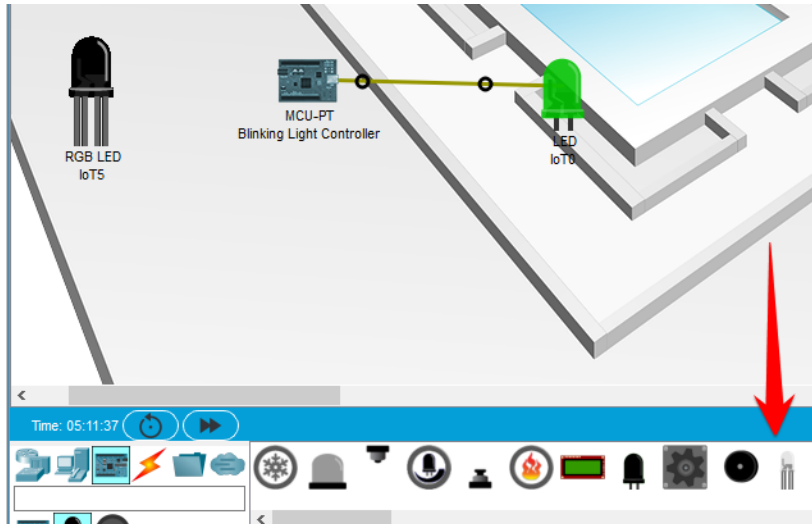
Part 2: Control a RGB LED using Blockly

In Part 2, you used Blockly to control an RGB LED. An RGB can display different colours with the combination of red, green, and blue.

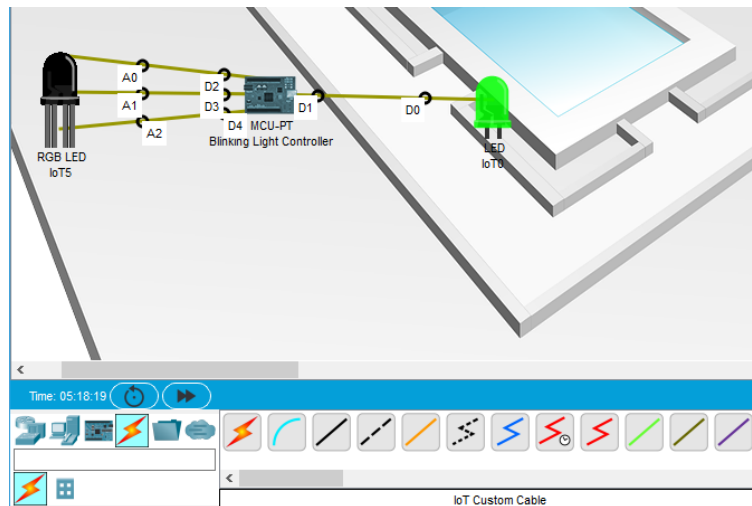
Step 1: Add an RGB LED.

In Step 1, you add a RGB LED into the workspace.

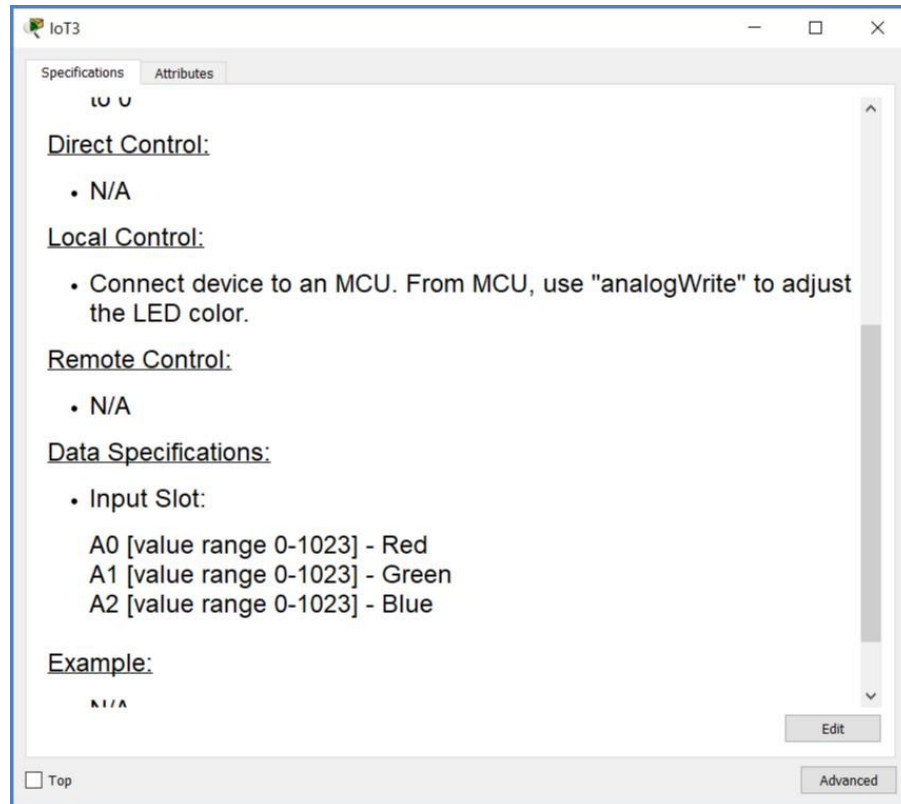
a) Click the subcategory **Actuators**, select **RGB LED** and drag it to the workspace.



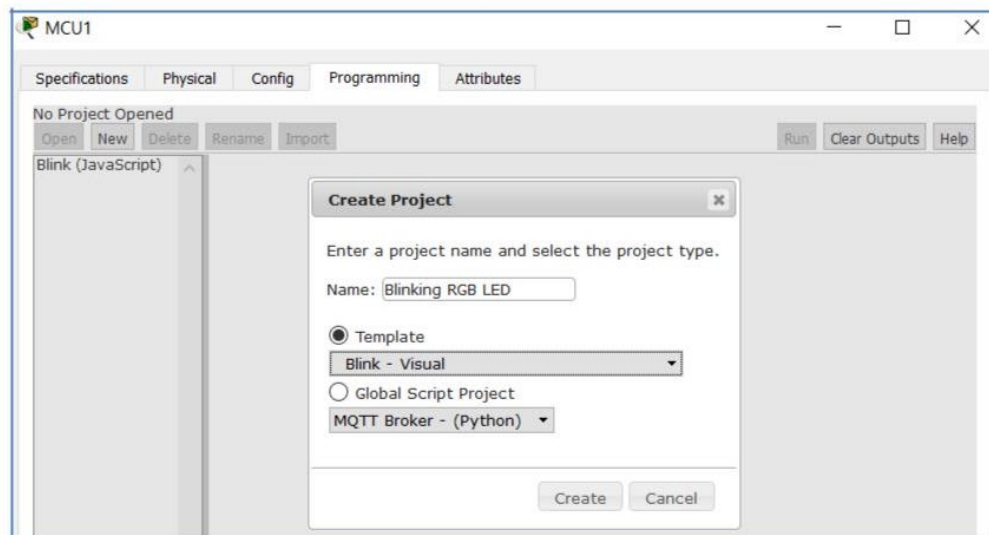
b) Click the **Connections** category, select three **IoT Custom Cables** to link MCU and RGB LED.



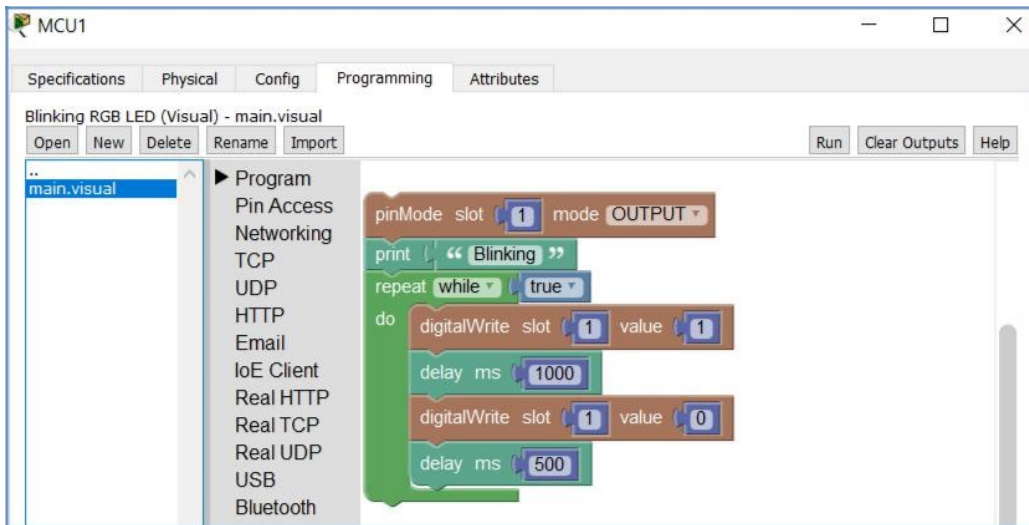
c) Click **RGB LED** and review its specification. Note different pin inputs represent different colours.



d) Open the prebuilt Blockly program. Click MCU -> Programming. Click on the '..' above 'main.visual', to move up a directory. This will allow you to create a new coding project, to configure the additional LED. Under the note **No Project Opened**, click **New**. In the **Create Project** window, enter **Blinking RGB LED** as the project name. In the Template dropdown menu, select **Blink - Visual**.



- e) Click **Create**. Double-click main.visual. The prebuilt Blockly program displays.



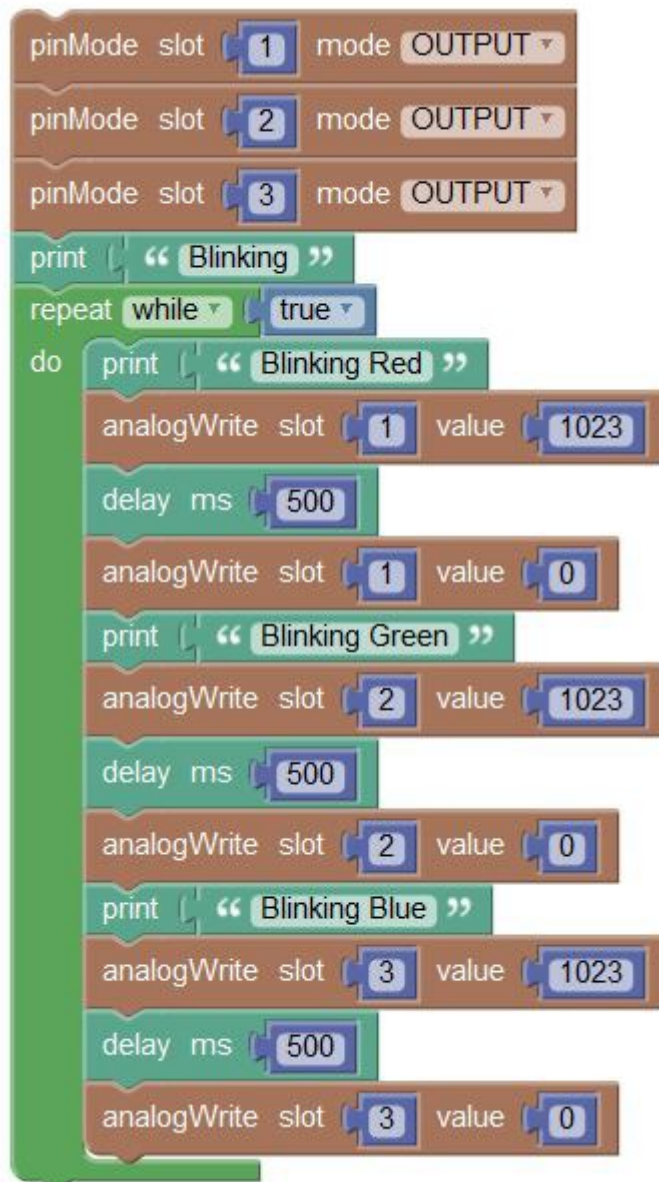
Step 2: Modify the Blockly program

- a. Expand **Pin Access** group, and add two more **pinMode** blocks to set three slots as **OUTPUT** (from MCB to send a signal to RGB LED). Set the slots to match where the cables were attached previously (when connecting the MCU to the LED)



- b. From the **Pin Access** group, select **analogWrite** blocks to replace **digitalWrite** blocks. Also, add a few **print** blocks.

The final program is as follows:



- d. Run the program. The LED should display RED, GREEN, and BLUE in sequence.

Challenge

Modify the program to show a combined colour from all three inputs with different, randomly generated values for each slot.

End of Practical 8